Sujan Dora

AI Engineer

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Professional Summary

AI/ML Engineer with 2 years of experience building LLM-RAG based document QA and production-ready AI solutions. Delivered systems that improved document QA accuracy by ∼20%, reduced MFA verification load by ∼25%, and optimized ID authentication pipelines, with hands-on experience in Python, PyTorch, TensorFlow, OpenCV, and AWS. Recent graduate with strong foundation in LLMs and RAG systems, model evaluation, and end-to-end ML workflows.

Education

Missouri State University — GPA: 3.63
M.S. Computer Science – Data Science — Dec 2025

Technical Skills

Programming Languages	Python, SQL
ML Frameworks & Libraries	TensorFlow, PyTorch, Scikit-learn, Hugging Face Transformers
LLM & NLP Tools	LangChain, LangGraph, LangSmith, LlamaIndex, Agents, Multi-Agents, LoRA, QLoRA, OpenAI, Ollama, RAG Systems
ML Specializations	NLP, Computer Vision, Deep Learning, Anomaly Detection
MLOps & Deployment	MLflow, Git, GitHub Actions, CI/CD Pipelines, Model Deployment
Cloud & Infrastructure	AWS (Lambda, S3, EC2, ECR, API Gateway)
Databases & AI Infrastructure	PostgreSQL, MySQL, Milvus, Weaviate, FAISS, MongoDB
Web & DevOps	FastAPI, Docker Containerization

Work Experience

AI/ML Engineer — Chennai, India
Grootan Technologies — July 2023 December 2023

- Implemented a self-correcting RAG system using LangChain and Mistral 7B, improving answer reliability and mitigating context errors in document-level QA.
- Built a hybrid retrieval workflow combining dense embeddings and BM25 search, enhancing relevance and coverage of retrieved documents.
- Developed an LLM-as-Judge evaluation pipeline to assess faithfulness and relevance across hundreds of QA examples, ensuring production-ready reliability.
- Introduced batched query extraction, reducing LLM calls by ∼80% per query and enabling faster, reliable production inference.
- Engineered production RAG system with self-correction mechanisms, hybrid retrieval (dense embeddings via FAISS/Milvus + BM25 sparse search), and LLM-as-Judge evaluation achieving ~20% accuracy improvement and 94%+ faithfulness scores for document QA workflows processing thousands of daily queries.
- Optimized LLM inference performance by implementing batched query extraction, caching strategies, and prompt versioning, reducing API calls by ~80% per query while maintaining sub-second latency for production servicescritical for cost-effective and responsive AI applications.
- Designed an ML-based anomaly detection system for MFA workflows, reducing redundant verification prompts by 25%, improving user experience while maintaining security coverage.

AI/ML Intern — Chennai, India
Grootan Technologies — September 2022 July 2023

- Built a National ID authentication system for detecting counterfeit holograms using YOLO object detection, MiDaS depth estimation, and GAN-generated synthetic samples to improve robustness.
- Optimized the hologram verification pipeline by eliminating redundant processing, reducing per-card detection time from 7 minutes to 3 minutes.
- Applied transfer learning to fine-tune pre-trained models for limited datasets, achieving reliable counterfeit detection across varied hologram designs.
- Containerized and deployed solutions using Docker and AWS, ensuring scalable production-ready pipelines for anomaly detection, identity verification, and document QA systems.

Data Science Intern — Bengaluru, India
iNeuron.ai — November 2021 March 2022

- Developed an end-to-end credit card default prediction project using Docker, CircleCI, and Heroku, achieving 87% model accuracy on simulated customer transactions.
- Built a CI/CD pipeline to deploy predictive models to AWS production, enabling real-time predictions.
- Conducted exploratory data analysis on 30,000+ customer records to identify key features that improved model performance.
- Implemented data validation and preprocessing pipelines, reducing manual data preparation effort.

Personal Projects

Wine Quality Prediction - End-to-End MLOps Pipeline

- Built an end-to-end ML pipeline using ZenML and MLflow, deploying a REST API on AWS Lambda with Docker for automated training and inference.
- Implemented CI/CD with GitHub Actions to automate model retraining and deployment to AWS (S3, ECR, API Gateway), ensuring reliable and repeatable releases.

BioMedical Research Assistant Multi-Agent LLM System

- Fine-tuned Llama-3.1-8B using QLoRA on PubMedQA data with 4-bit quantization, gaining hands-on experience in parameter-efficient LLM training for biomedical QA.
- Built a multi-agent research workflow using CrewAI and PubMed retrieval via Model Context Protocol, with a Streamlit interface for interactive querying.

SafeSpace 2.0 AI Mental Health Therapist

- Developed an AI-powered mental health companion with empathetic conversational support, crisis detection, and location-aware therapist recommendations.
- Implemented a tool-using LLM agent (LangGraph / REAct) with multi-channel frontends (Streamlit web chat + Twilio WhatsApp) and integrated APIs for real-time actionable outputs.

Publications & Research

Dora, S. et al. (2021). "Detecting Bias in the Relative Grading System Using Machine Learning." *IEEE International Conference on Computing and Communications Technologies.*